



13.0.1 ISRO CSE 2009 | Question: 49



$x = a \cos(t), y = b \sin(t)$ is the parametric form of

- A. Ellipse B. Hyperbola C. Circle D. Parabola

isro2009 calculus

Answer key

13.1

Differentiation (1)

13.1.1 Differentiation: ISRO CSE 2011 | Question: 59



n -th derivative of x^n is

- A. nx^{n-1} B. $n^n \cdot n!$ C. $nx^n!$ D. $n!$

isro2011 calculus differentiation

Answer key

13.2

Extreme Points (1)

13.2.1 Extreme Points: ISRO-DEC2017-4



If $x = -1$ and $x = 2$ are extreme points of $f(x) = \alpha \log |x| + \beta x^2 + x$ then

- A. $\alpha = -6, \beta = \frac{-1}{2}$ B. $\alpha = 2, \beta = \frac{-1}{2}$
C. $\alpha = 2, \beta = \frac{1}{2}$ D. $\alpha = -6, \beta = \frac{1}{2}$

isrodec2017 calculus extreme-points

Answer key

13.3

Functions (1)

13.3.1 Functions: ISRO CSE 2018 | Question: 15



The domain of the function $\log(\log \sin(x))$ is:

- a. $0 < x < \pi$ b. $2n\pi < x < (2n+1)\pi$, for n in N
c. Empty set d. None of the above

isro2018 calculus functions

Answer key

13.4

Limits (1)

13.4.1 Limits: ISRO CSE 2016 | Question: 3



$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$ is given by

- A. 0 B. -1 C. 1 D. $\frac{1}{2}$

Answer key 

13.5

Maxima Minima (2)

13.5.1 Maxima Minima: ISRO CSE 2009 | Question: 50



The value of x at which y is minimum for $y = x^2 - 3x + 1$ is

- A. $-3/2$ B. $3/2$ C. 0 D. $-5/4$

isro2009 calculus maxima-minima

Answer key 

13.5.2 Maxima Minima: ISRO-2013-49



What is the least value of the function $f(x) = 2x^2 - 8x - 3$ in the interval $[0, 5]$?

- A. -15 B. 7 C. -11 D. -3

isro2013 maxima-minima

Answer key 

13.6

Vector Calculus (1)

13.6.1 Vector Calculus: ISRO CSE 2009 | Question: 45



Which of the following statement is correct

- A. $\Delta(U_k V_k) = U_k \Delta V_k + V_k \Delta U_k$
 B. $\Delta(U_k V_k) = U_{k+1} \Delta V_k + V_{k+1} \Delta U_k$
 C. $\Delta(U_k V_k) = V_{k+1} \Delta U_k + U_k \Delta V_k$
 D. $\Delta(U_k V_k) = U_{k+1} \Delta V_k + V_k \Delta U_k$

isro2009 calculus vector-calculus non-gatecse

Answer key 

Answer Keys

13.0.1	A	13.1.1	D	13.2.1	B	13.3.1	C	13.4.1	C
13.5.1	B	13.5.2	C	13.6.1	B				